

Oscillator



[An Oscillator is a device that incorporates active and passive circuit elements to generate sinusoidal or other repetitive waveforms of nearly or constant frequency]

OSCILLATOR

Requirements of Oscillation (Barkhausen's Criterion)

In practical it is not sufficient just to make the equation:

$$\beta A = 1$$

$$A' = \frac{A}{1 + A\beta}$$

$A\beta$ is made larger than 1 and if it is negative, the above equation does not hold any longer since the circuit is no longer an amplifier.

Since β can not usually be greater than 1,

$A\beta \geq 1$ means that

$$A \geq \frac{1}{\beta}$$

Suppose that 1.5% of the output of an amplifier is fed back positively to the input. Then minimum gain (A) required of the amplifier for Oscillation to occur is:

$$A \geq \frac{1}{\beta} = \frac{1}{0.015} = \frac{1000}{15} = 66.67$$

Thus if $A = 100$ and $\beta = 0.5\%$ (0.005)

$$A\beta = 0.5$$

so that the ckt. can not Oscillate.

If it is increased to 2% ($\beta = 0.02\%$)

$A\beta = 2$, Oscillations will now take place

[Example Given above:

$$A_v = 100 \quad \text{and} \quad \beta = 0.5\%$$

$$\text{Now, } A_v \geq \frac{1}{\beta} = \frac{1}{0.005} = \frac{1000}{5} = 200.$$

$$\therefore A\beta = 100 \times 0.005 = 0.5 \quad \text{i.e. } A\beta < 1$$

Hence this circuit can not Oscillate.



Oscillator

But if $\beta = 0.02$ (2%)

$$\therefore A\beta = 100 \times 0.02 = 2$$

$$\therefore A\beta > 1$$

Hence the circuit is now Oscillate.]

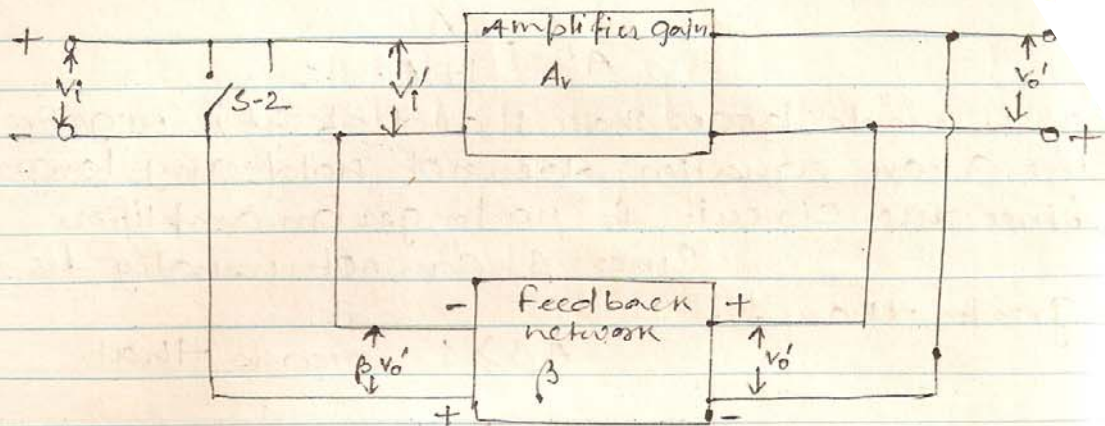


Fig: 1: Block diagram of an Oscillator (osc) in which both A_v and β are negative.

Consider the ckt. in Fig:1 in which the amplifier causes a phase shift of 180° between V_i & V_o . In order to provide (+ve) feedback, the feedback network must (a) provide a phase shift of 180° so as to give a signal that can replace V_i and

(b) not cause more attenuation than $\frac{1}{A_v}$, since βA_v must be equal to or greater than unity.

This condition of unity loop gain ($A\beta=1$) is known as the "Barkhausen Criterion" for Oscillator and may be met by β and A_v being both (+ve) or both (-ve). When this condition is satisfied and V_i removed and the output will continue.

Thus we find that if in a positive feedback amplifier the loop gain is greater than unity instabilities set in and the circuit tends to Oscillate. The oscillations are sustained when $A\beta = 1$.



the requirement for oscillation, $\beta A_v \geq 1$ is not met at only one frequency, sinusoidal oscillations will result and we have a "Sine-wave Oscillator".

If the requirement is met over a band of frequencies, then the output will be non-sinusoidal in wave form such as (a) Sawtooth, square wave etc.

(1) The Phase-Shift Oscillator (R-c)

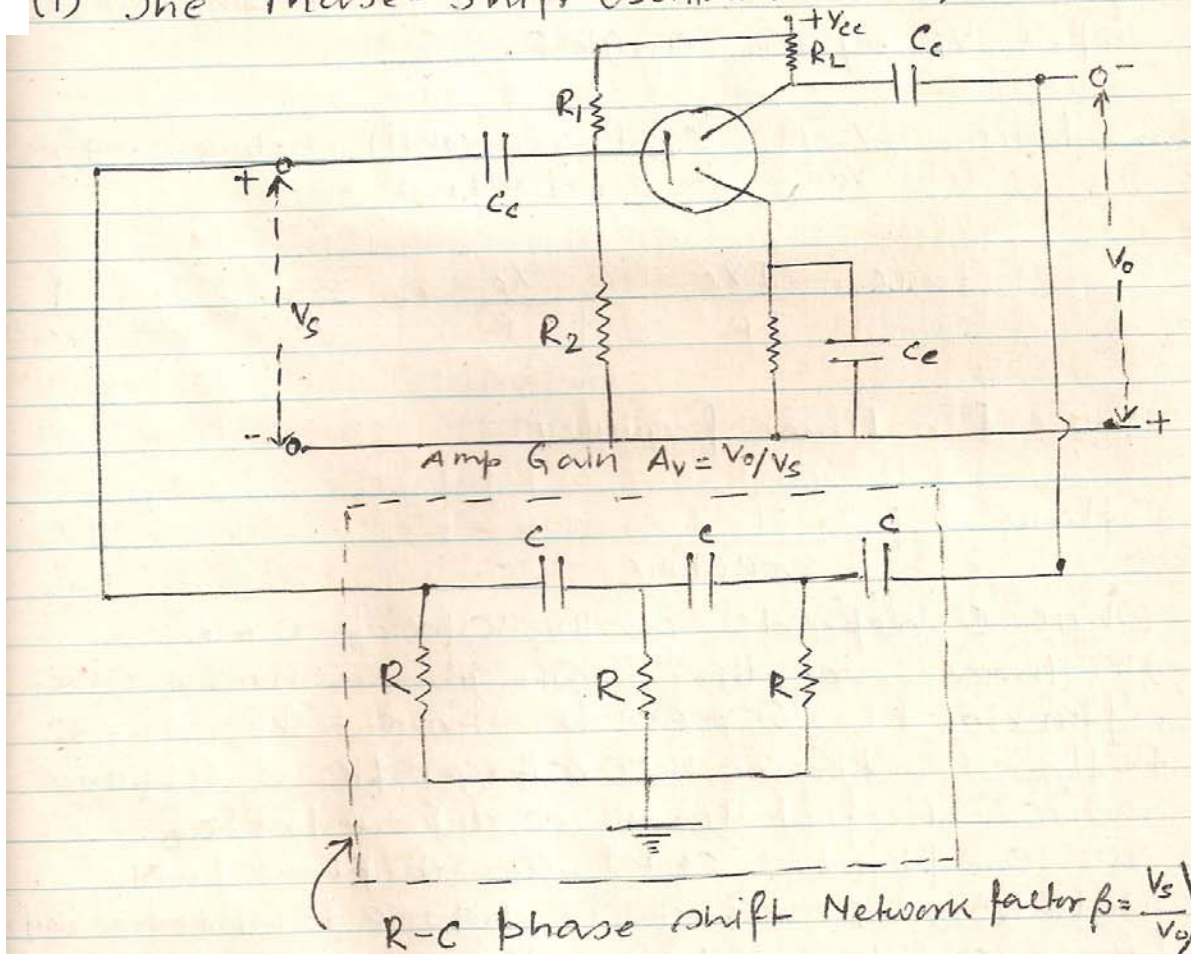
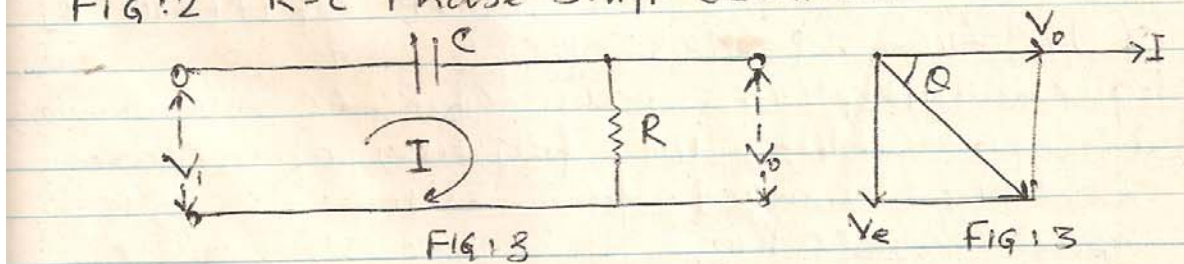


FIG:2 R-c Phase Shift Oscillator.





The Cir. Shown in fig:2 Consists of an amp stage and a feed-back network to provide an additional phase shift of 180° in the output signal. Now, from fig-3

$$V_o = IR$$

$$V_c = IX_c$$

and the voltage V_o across R is in phase with the current I but it is leading with input V_i by an angle θ .

$$\tan \theta = \frac{V_c}{V_o} \quad \therefore \text{Phase shift between input and output is}$$

$$\tan \theta = \frac{IX_c}{IR} = \frac{X_c}{R} = \frac{\frac{1}{\omega C}}{R} = \frac{1}{\omega CR} = \frac{1}{2\pi fCR}$$

where f is linear frequency.

$$\text{Hence } f = \frac{1}{2\pi RC \tan \theta}$$

Thus θ depends on the value of C & R . If there are three stages each must give approximately $\theta = 60^\circ$ i.e. $\tan \theta = \sqrt{3} = 1.732$. Hence the voltage after 3-sections $f = \frac{1}{2\pi \sqrt{3} RC}$ which is fed back to the input of the amplifier would be in phase with input signal, since it has a total phase shift of 180° . [for example ; if $R = 220 \text{ k}\Omega = 220 \times 10^3 \Omega$ & $C = 500 \text{ pF} \quad | \quad 1 \text{ pF} = 10^{-12} \text{ F}$]

of 180° then $f = 1835 \text{ Hz}$. by changing the value of R & C simultaneously in each section, the frequency of oscillation can be changed from as low as 20 Hz to as much as 300 kHz .



In the above (example) discussion we are ignored the additional current that flows through C for the other sections.

It is to be noted that there is only one frequency at which the overall (net) phase shift around the loop will be 0° or 360° so that a sine wave output is obtained. This output will be free from distortion if the gain of the amplifier slightly exceeds the loss in R-C Network to make the loop gain a little over 1.

Since it is known that a 3-section network causes the output to be $1/29$ of the input, the gain of the amplifier must exceed 29 and the transistor must be chosen accordingly. The phase shift Oscillator are commonly used for audio-frequency ranges. (low).

Working: A transistor phase shift Oscillator is shown in fig: 2. The resistors R_1, R_2, R_L, R_E and the Oscillator power supply V_{CC} serves to establish the d.c. operating point of the transistor in the active region of its characteristic. C_E is an emitter bypass capacitor.

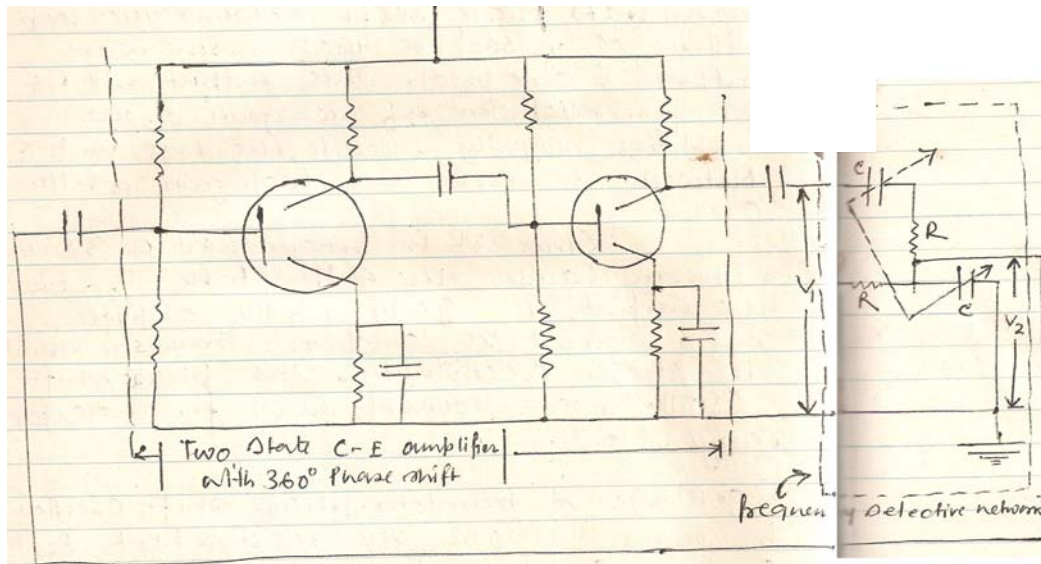
The transistor operates in the CE mode and so produces a phase shift of 180° between the input and output voltages. An additional 180° shift is produced by the three sections of R-C network.



Oscillator

(2) Wien - Bridge Oscillator :

This is the Oscillator which can generate readily a variable frequency output. The circuit is shown below.



The two stage common emitter amplifier provide a phase shift of 360° (approx) so that the frequency selective network has not to introduce any additional phase shift.

The conditions for V_2 to be in phase with V_1 are met with when $X_C = R$ or $f = \frac{1}{2\pi RC}$. Under this conditions $V_2 = \frac{V_1}{3}$ and the attenuation is not as high as in the phase shift oscillator network.

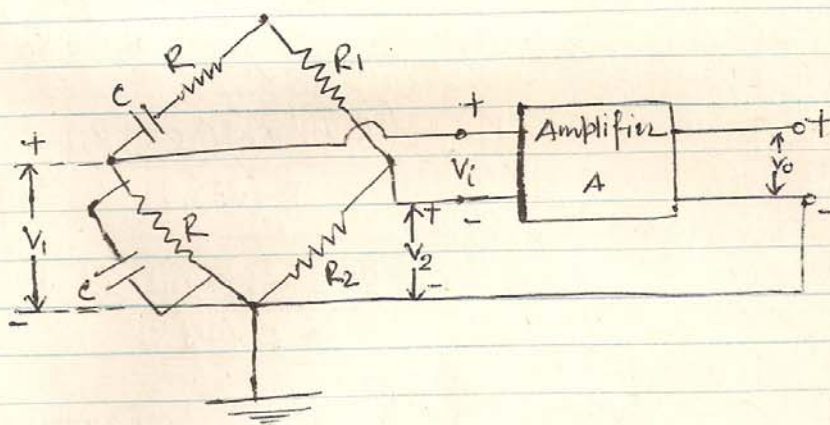
Variable frequency operation requires the simultaneous adjustment of



two capacitors, an easier task than the phase shift oscillator. This bridge oscillator is capable of generating frequency from 20 Hz to 200 kHz with a distortion of less than 1%.

Wien - Bridge Oscillator:

The Wien bridge oscillator is an RC oscillator in which a (nearly) balanced bridge is used as the feedback network as shown in fig: 5.



The output voltage (V_o) of the amplifier is the input voltage of the bridge and the output voltage of the bridge (V_i) is fed to the input of the amplifier.

At exact balance of the bridge, the output voltage and the bridge V_i is zero. Hence the feed-back fraction

$$\beta = \frac{V_i}{V_o} = 0$$

But for sustained oscillations, β must be non-zero to make the loop gain equal to unity.

This condition is achieved by slightly unbalancing the bridge by changing the ratio



between the resistances R_1 and R_2 . The bridge introduces no phase shift between its input and output voltages. The amplifier should not introduce any phase shift between its input and output voltages, so that the net phase shift around the loop is zero. A two stage C-E transistor amplifier may be used for this purpose.

Analysis:

At balance we can write:

$$\begin{aligned} \frac{R_1}{R_2} &= \frac{\left(R + \frac{1}{j\omega C}\right)}{\left\{ \frac{R(-jX_C)}{R - jX_C} \right\}} \\ &= \frac{(R - jX_C)(R - jX_C)}{R(-jX_C)} = \frac{R^2 - 2jRX_C + j^2X_C^2}{R(-jX_C)} \\ &= \frac{R^2 - 2jRX_C - X_C^2}{R(-jX_C)} \\ &= \frac{R^2 - X_C^2}{R(-jX_C)} - \frac{2jRX_C}{R(-jX_C)} \end{aligned}$$

$$\therefore \frac{R_1}{R_2} = j \frac{R^2 - X_C^2}{RX_C} + 2$$

Equating the real and imaginary parts we get

$$\frac{R_1}{R_2} = 2 \quad \text{or} \quad R_1 = 2R_2$$

$$\text{and } R^2 - X_C^2 = 0$$

$$\text{As } R^2 - X_C^2 = 0$$

$$X_C^2 = R^2$$

$$\frac{1}{\omega C} = R$$

$$\frac{1}{2\pi f C} = R$$

$$\text{or } f_0 = \frac{1}{2\pi RC}$$

where f_0 is the frequency of oscillations.



Oscillator

EXPRESSION FOR GAIN IN LOW AND MEDIUM FREQUENCIES :

We have, $|A| = \frac{V_o}{V_s}$

Where, V_o = Output potential and
 V_s = Signal

Then from eqn,

$$V_o = \left(\frac{V_s}{\text{Total resistance}} \right) \times R_i$$

$$= \frac{V_s}{\left(R_i + R_s - \frac{j}{\omega C} \right)} \times R_i$$

$$\therefore \frac{V_o}{V_s} = \frac{R_i}{\left(R_i + R_s - \frac{j}{\omega C} \right)}$$

$$= \frac{R_i}{(R_i + R_s)} \left\{ \frac{1}{1 - \frac{j}{\omega C(R_i + R_s)}} \right\} \quad \text{--- (1)}$$

$\rightarrow + \frac{1}{j\omega C} = - \frac{j}{\omega C}$
 C = Capacitance for C_c .

Let

ω_L = Angular frequency denoting low and middle range frequency.

So that $f_L = \frac{1}{2\pi} \cdot \omega_L$

and $\omega_L = \frac{1}{C \cdot (R_i + R_s)}$

So that equation (1) reduces to

$$\frac{V_o}{V_s} = \frac{R_i}{(R_i + R_s)} \left\{ \frac{1}{1 - \frac{j\omega_L}{\omega}} \right\} \quad \text{--- (2)}$$

Then from eqn (2): $|A| = \frac{V_o}{V_s} = \frac{R_i}{(R_i + R_s)} \left\{ \frac{1}{1 - \frac{j\omega_L}{\omega}} \right\}$ --- (3)

Now, at $\omega_L = \omega$ eqn (3) reduces to

$$|A| = \frac{R_i}{(R_i + R_s)} \left\{ \frac{1}{1 - j} \right\} \quad \text{--- (4)}$$



Oscillator

Value of $\left(\frac{1}{1-j}\right)$:

$$\text{Now, } \frac{1}{1-j} = \frac{1+j}{(1-j)(1+j)} = \frac{1+j}{1-j^2} = \frac{1+j}{2}$$

$$\text{But } |1+j| = \sqrt{1^2+1^2} = \sqrt{2}$$

$$\therefore \frac{1}{1-j} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

This equation (4) becomes: $|A| = \frac{R_i}{R_i+R_s} \left\{ \frac{1}{\sqrt{2}} \right\}$

Whence,

$$|A| = 0.707 \left(\frac{R_i}{R_i+R_s} \right) \quad \text{--- (5)}$$

The variation of "gain" with frequency is shown below

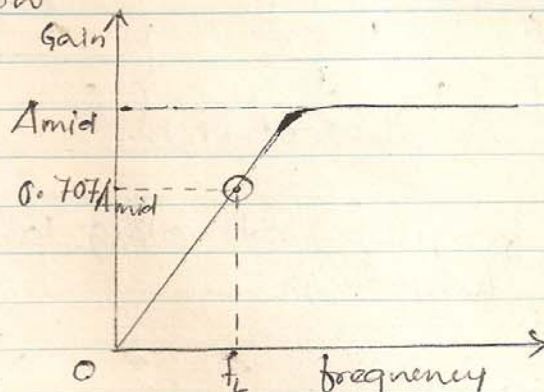


FIG:2

The mid-band-voltage-gain when C_c is a "short" is given by,

$$A_{mid} = \frac{R_i}{R_i+R_s} \quad \text{--- (6)} \quad \left\{ \begin{array}{l} \text{from (1)} \\ \text{because} \\ \frac{1}{j\omega C_c} = 0 \end{array} \right.$$

Hence, the magnitude of the gain from eqⁿ (5) & (6) appears to fall-off to 70.7% of the mid-band value as shown in fig:2 at $f = f_L$. This frequency (f_L) is called the "Lower cutoff frequency" which is lower limit for the amplifier frequency response.

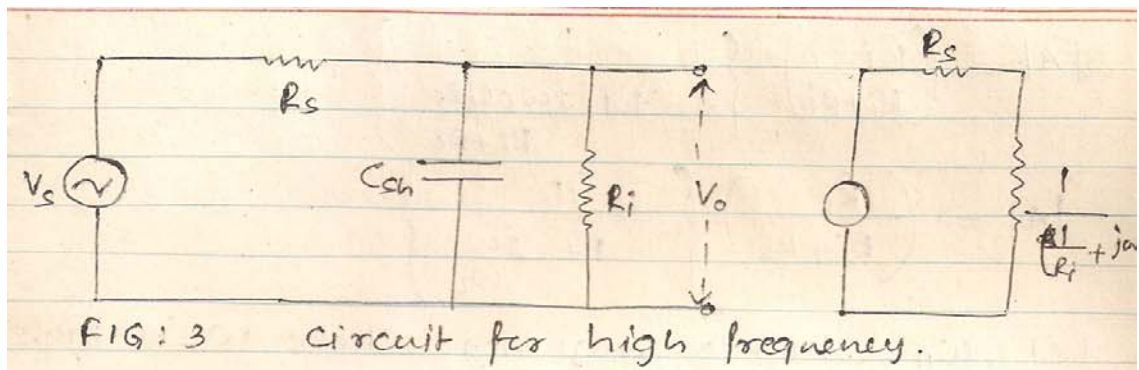


Fig: 3 shows the equivalent circuit at high frequency. At low and mid-frequency the Capacitor C_c which is of low value will offer high resistance ($\frac{1}{\omega C_c}$). As the frequency is increased the reactance ($\frac{1}{\omega C_c}$) of C_{sh} decreases and begins to decrease the impedance across which V_o is produced. More voltage is therefore dropped over R_s and less is available at the output. Thus the voltage gain of the amplifier decreases at high frequency and the circuit behaves as a "LOW PASS Filter".

The output voltage is given by :

$$V_o = \frac{V_s}{(R_s + \dots)} \times \dots$$

$$V_o = \frac{V_s}{\left\{ R_s + \frac{R_i}{(1 + j\omega C R_i)} \right\}} \times \left\{ \frac{R_i}{(1 + j\omega C R_i)} \right\}$$

$$\begin{aligned} \text{or } \frac{V_o}{V_s} &= \left(\frac{R_i}{1 + j\omega C R_i} \right) \left[\frac{1}{\frac{1}{(1 + j\omega C R_i)} \{ R_i + R_s(1 + j\omega C R_i) \}} \right] \\ &= \frac{R_i}{(1 + j\omega C R_i)} \left\{ \frac{(1 + j\omega C R_i)}{R_i + R_s(1 + j\omega C R_i)} \right\} \end{aligned}$$

$$\therefore |A| = \left\{ \frac{R_i}{R_i + R_s + j\omega C R_i R_s} \right\} \quad \text{--- (7)}$$



Oscillator

$$|A| = \left(\frac{R_i}{R_i + R_s} \right) \left\{ \frac{1}{1 + \frac{j\omega R_i R_s}{R_i + R_s}} \right\}$$

$$|A| = \left(\frac{R_i}{R_i + R_s} \right) \left\{ \frac{1}{1 + \frac{j\omega}{\omega_H}} \right\}$$

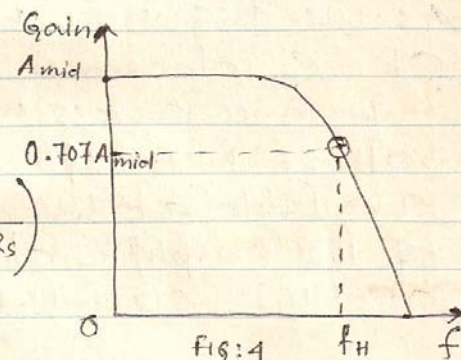
Let ω_H = Angular frequency where ω is high then,

$$|A| = \left(\frac{R_i}{R_i + R_s} \right) \left(\frac{1}{1 + \frac{j\omega}{\omega_H}} \right)$$

Where,

$$\omega_H = \frac{R_i + R_s}{C_{sh} R_i R_s}$$

So that $f_H = \frac{1}{2\pi} \left(\frac{R_i + R_s}{C_{sh} R_i R_s} \right)$



At $\omega = \omega_H$

$$\therefore \frac{V_o}{V_s} = \left(\frac{R_i}{R_i + R_s} \right) \frac{1}{(1 + j)}$$

$$= \left(\frac{R_i}{R_i + R_s} \right) \frac{1}{\sqrt{2}} \quad \therefore \text{Hence, } |A| = 0.707 A_{mid}$$

Thus as a combine effect due to Series and Shunt Capacitors, the amplifier behaves as a band pass-amplifier amplifying the frequencies between the two cut-off frequency f_L and f_H as shown in fig: 5. At these two frequencies f_L and f_H the voltage gain falls to $\frac{1}{\sqrt{2}}$ of the mid-range voltage gain. These frequencies are therefore also called "Half-Power frequency".

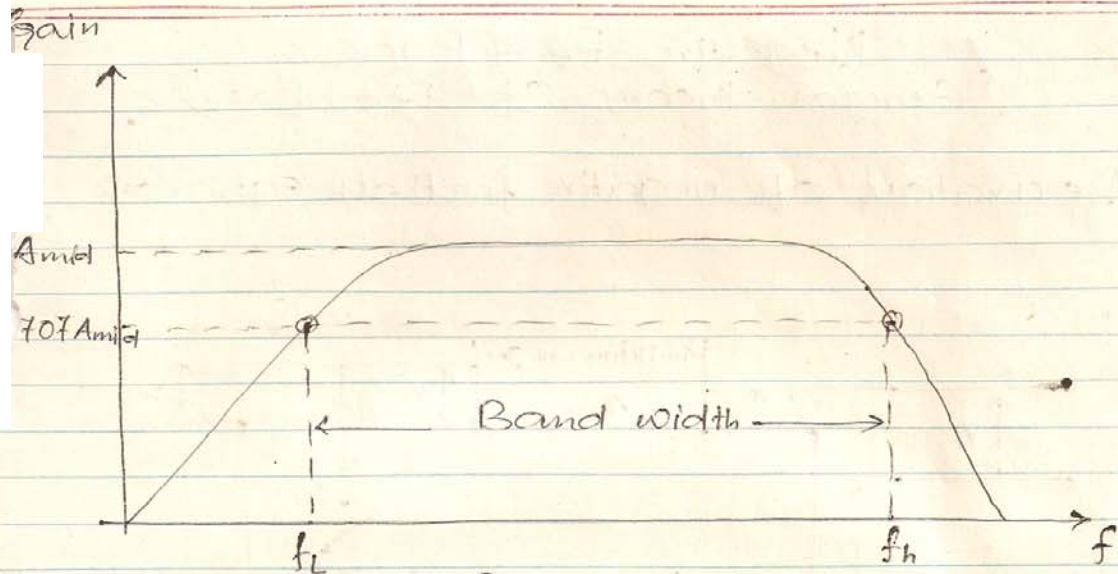


FIG: 5

Just as gain varies with frequency, the phase shift caused by the amplifier also varies with frequency. This phenomenon is known as "Phase Distortion".

The phase shift is defined in terms of an angle θ which is the shift in phase for each Fourier component of the signal as it passes through the amplifier. It is the shift in phase measured in second multiplied by the ang. frequency of the Fourier component.

To avoid the phase distortion suitably designed phase shifting networks are added at the output of the amplifier.

For the "high pass" and "low pass" circuit discuss earlier (fig: 3 & fig: 4), the relative phase shift w.r to mid-band are as follows

(a) At low frequency : $\theta = + \tan^{-1} \frac{f_L}{f}$

(b) At high frequency $\theta = - \tan^{-1} \frac{f}{f_H}$

At $f = f_L$ and $f = f_H$ the relative phase shift is $+45^\circ$ and -45° respectively. At extremely low frequency it will be $+90^\circ$ and at high frequency -90° . Hence a high-pass circuit is (fig-1) called a "Lead Network" and the low-pass circuit (fig-3) is known as "LAG Network".